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RESEARCH SUMMARY

Precipitation retrieval and product-evaluation scientist specializing in satellite algorithm development, diagnostic evaluation, uncertainty analysis, and geospatial machine learning. My research focuses on improving precipitation measurement by integrating infrared and microwave remote sensing, in situ observations, re-analysis data, physical correction methods, and machine-learning retrieval approaches. At the University of Arizona, I contribute to NASA-supported GPCP/IMERG-relevant high-latitude precipitation research, including AVHRR-based retrieval development, limb-darkening correction, diagnostic evaluation of satellite and reanalysis precipitation products, and multi-reference validation over ocean and polar regimes. My doctoral work developed methods for combining Meteosat satellite observations with commercial microwave link rainfall signals for rainfall detection and mapping in data-sparse regions.

PROFESSIONAL APPOINTMENTS

Postdoctoral Research Associate

Department of Hydrology & Atmospheric Sciences, University of Arizona, Tucson, AZ
Behrangi

2023–Present

Supervisor: Prof. Ali

NASA-supported research context: GPCP high-latitude precipitation improvement, IMERG-relevant high-latitude precipitation assessment, and precipitation-product evaluation.

- Key technical contributor to the development, evaluation, and production-style delivery of UA-HIPA, an AVHRR-based precipitation retrieval product for regions poleward of 55°; developed a physically informed limb-darkening correction for AVHRR infrared brightness temperatures to address cross-swath scan-geometry bias.
- Built diagnostic evaluation tools to assess AVHRR-based precipitation estimates across latitude bands, seasons, surface types, and independent precipitation references.
- Contributed to GPCP Version 3.3 diagnostic evaluation and coauthored the submitted Journal of Hydrometeorology manuscript JHM-D-26-0048 with members of the GPCP science team.
- Developing the Integrated Evaluation of Precipitation Products over Oceans (IEPPO) framework, comparing GPCP, IMERG, ERA5, and MERRA-2 against PAL, moored buoy, atoll, and OceanRAIN references across daily to climatological timescales.
- Lead analyst for an Antarctic precipitation benchmarking study using GRACE-based mass-budget constraints, ice-discharge estimates, and sublimation inputs to evaluate ERA5, GPCP v3.3, and GPM constellation precipitation estimates across IMBIE drainage basins.

Visiting Scientist – EUMETSAT / H-SAF

European Organisation for the Exploitation of Meteorological Satellites, Darmstadt, Germany

2022–2023

- Conducted precipitation-product evaluation through EUMETSAT's Satellite Application Facility on Support to Operational Hydrology and Water Management, with emphasis on satellite precipitation products over Sub-Saharan Africa.
- Contributed to diagnostic evaluation, uncertainty characterization, and validation of operational precipitation products.

Doctoral Researcher

Department of Water Resources, University of Twente, Enschede, Netherlands

2016–2022

- Developed methods for high-spatiotemporal-resolution rainfall estimation by combining Meteosat Second Generation (MSG) SEVIRI cloud-top information with commercial microwave link rainfall observations.
- Built and evaluated rainfall-detection, rain-area identification, and gridded precipitation-estimation methods for data-sparse regions in Sub-Saharan Africa.
- Published peer-reviewed articles on CML-satellite fusion, satellite-supported rain-area detection, and near-real-time rainfall mapping.

EDUCATION

- Ph.D., GIS & Earth Observation (Hydrometeorology / Water Resources)** 2022
University of Twente, Faculty ITC, Enschede, Netherlands
Dissertation: “High-Spatiotemporal Resolution Rainfall Estimation from Satellite and Commercial Microwave Link Data.” Supervisors: Prof. Bob Su (ITC/TU Delft); Co-supervisor: Prof. Ben Maathuis.
- M.Sc., GIS & Earth Observation (Water Resources & Environmental Management)** 2016
University of Twente, Faculty ITC, Enschede, Netherlands
University of Twente ITC Excellence Scholarship (UTS-ITC), €20,000 – competitive merit-based funding awarded by the Faculty ITC to internationally selected candidates.
- B.Sc., Environmental Science** 2012
University of Cape Coast, Ghana

SELECTED SCIENTIFIC CONTRIBUTIONS

UA-HIPA – AVHRR High-Latitude Precipitation Retrieval

Key technical contributor to the University of Arizona–High-latitude Infrared-based Precipitation Analysis Version 2 (UA-HIPA), an AVHRR-based precipitation retrieval product for global regions poleward of 55°. The product uses AVHRR observations from NOAA and MetOp polar-orbiting satellites, reanalysis fields, and machine-learning retrieval methods to estimate rain and snow in high-latitude regions where conventional satellite sampling remains limited.

AVHRR Limb-Darkening Correction Algorithm

Developed and implemented a physically informed limb-darkening correction for AVHRR infrared brightness temperatures, addressing cross-swath scan-geometry bias in the input data used for high-latitude precipitation retrieval. The correction framework is conditioned on latitude, season, cloud type, surface type, and scan geometry.

CML-Satellite Rainfall Retrieval and External Implementation

Developed peer-reviewed methods combining commercial microwave link attenuation with MSG SEVIRI satellite observations for rainfall detection, estimation, and gridded rainfall mapping in data-sparse regions. This research line later attracted external implementation interest for African rainfall-monitoring applications through Rainboo/TAHMO/TU Delft-related prototyping.

GMASI Snow and Ice Cover Record Extension, 1980–1987

Developed and validated machine-learning and climatology-based methods using ERA5-derived surface variables to extend the Global Merged Analysis of Snow and Ice record back to 1980–1987. The work was published in *Earth and Space Science*, with the resulting dataset deposited in the University of Arizona Research Data Repository.

IEPPO – Multi-Reference Ocean Precipitation Validation Framework

Developing an integrated, sampling-aware ocean precipitation evaluation framework comparing satellite and reanalysis products, including GPCP, IMERG, ERA5, and MERRA-2, against complementary in situ references such as PAL, moored buoys, coral atolls, and OceanRAIN.

Antarctic Snowfall Mass-Budget Benchmarking

Lead analyst for a physically constrained reassessment of Antarctic precipitation across IMBIE drainage basins using GRACE-based mass-storage change, ice discharge, basal melt, and sublimation inputs to evaluate ERA5, GPCP v3.3, and GPM constellation precipitation estimates.

SELECTED PEER-REVIEWED PUBLICATIONS & DATA PRODUCTS

- Kumah, K. K.,** Zandi, O., & Behrangi, A. (2025). Retrospective Mapping of Global Snow and Ice Cover Beyond the Satellite Observational Era. *Earth and Space Science*. <https://doi.org/10.1029/2024ea004171>
- Kumah, K. K.,** Maathuis, B. H. P., Hoedjes, J. C. B., & Su, Z. (2022). Near real-time estimation of high spatiotemporal resolution rainfall from cloud top properties of the MSG satellite and CML rainfall intensities. *Atmospheric Research*, 279. <https://doi.org/10.1016/j.atmosres.2022.106357>
- Kumah, K. K.,** Hoedjes, J. C. B., David, N., Maathuis, B. H. P., Gao, H. O., & Su, B. Z. (2021). The MSG Technique: Improving Commercial Microwave Link Rainfall Intensity by Using Rain Area Detection from Meteosat Sec-

ond Generation. *Remote Sensing*, 13(16). <https://doi.org/10.3390/rs13163274>

Kumah, K. K., Maathuis, B. H. P., Hoedjes, J. C. B., Rwasoka, D. T., Retsios, B. V., & Su, B. Z. (2021). Rain Area Detection in South-Western Kenya by Using Multispectral Satellite Data from Meteosat Second Generation. *Sensors*, 21(10). <https://doi.org/10.3390/s21103547>

David, N., Liu, Y., **Kumah, K. K.**, Hoedjes, J. C. B., Su, B. Z., & Gao, H. O. (2021). On the Power of Microwave Communication Data to Monitor Rain for Agricultural Needs in Africa. *Water*, 13(5), 730. <https://doi.org/10.3390/w13050730>

Kumah, K. K., Hoedjes, J. C. B., David, N., Maathuis, B. H. P., Gao, H. O., & Su, B. Z. (2020). Combining MWL and MSG SEVIRI Satellite Signals for Rainfall Detection and Estimation. *Atmosphere*, 11(9). <https://doi.org/10.3390/atmos11091688>

Miyittah, M. K., Tsyawo, F. W., & **Kumah, K. K.** (2015). Suitability of two methods for determination of point of zero charge (PZC) of adsorbents in soils. *Communications in Soil Science and Plant Analysis*.

Submitted / In Preparation

Behrangi, A., **Kumah, K. K.**, et al. (submitted 2026). GPCP Version 3.3: A Refined Global View of Precipitation from Satellite and Gauge Observations. *Journal of Hydrometeorology*, MS# JHM-D-26-0048.

Zandi, O., Behrangi, A., **Kumah, K. K.**, et al. (in preparation). University of Arizona–High-latitude Infrared-based Precipitation Analysis Version 2 (UA-HIPA): Algorithm development, evaluation, validation, and dataset documentation for GPCP/IMERG high-latitude precipitation assessment.

Kumah, K. K., et al. (in preparation). An Integrated Multi-reference Assessment of Ocean Precipitation Products with PAL, Buoys, Atolls, and OceanRAIN.

Kumah, K. K., et al. (in preparation). Revisiting Antarctic Snowfall Using a Mass-Budget Approach: Evaluation Against ERA5, GPCP v3.3, and GPM Constellation Satellites.

Open Dataset

Kumah, K. K., Zandi, O., & Behrangi, A. (2024). Global Snow and Ice Cover (1980–1987): Extended GMASI Dataset. University of Arizona Research Data Repository. <https://doi.org/10.25422/azu.data.28012532.v1>

SELECTED CONFERENCE PRESENTATIONS

EGU General Assembly 2026 – Vienna, Austria. Behrangi, A., G. Huffman, R. F. Adler, Y. Song, **K. K. Kumah**, D. T. Bolvin, E. J. Nelkin, & G. Gu. The Newly Released Global Precipitation Climatology Project (GPCP) V3.3 Daily and Monthly Products and the Future Plans. EGU26-5885.

AGU Fall Meeting 2025 – New Orleans, Louisiana. Zandi, O., S. Pfreundschuh, **K. K. Kumah**, & A. Behrangi. University of Arizona–High-latitude Infrared-based Precipitation Analysis Version 2 (UA-HIPA).

AGU Fall Meeting 2025 – New Orleans, Louisiana. Behrangi, A., G. Huffman, R. F. Adler, Y. Song, **K. K. Kumah**, D. T. Bolvin, E. J. Nelkin, & G. Gu. The Newly Released GPCP V3.3 Daily and Monthly Products and the Future Plans.

AGU Fall Meeting 2024 – Washington, D.C. **Kumah, K. K.**, Zandi, O., & Behrangi, A. On the Retrospective Mapping of Global Snow and Ice Cover Beyond the Satellite Observational Era. Session: Cryosphere / Artificial Intelligence.

AGU Fall Meeting 2024 – Washington, D.C. Zandi, O., **Kumah, K. K.**, & Behrangi, A. Developing a Novel Long-Term IR-based Precipitation Product from AVHRR in High Latitudes Through a Double Machine Learning Strategy. AGU Fall Meeting Abstracts 2024, H13U-04.

EUMETSAT Meteorological Satellite Conference 2022 – Brussels, Belgium. **Kumah, K. K.** High Spatial and Temporal Resolution Rainfall Estimation from Cloud Top Properties of MSG Using Random Forest Algorithm Trained with Microwave-Link-Derived Rainfall. Accepted oral presentation, Paper #107, Session S07-B.

KIT/HyMet CML Symposium 2019 – Garmisch-Partenkirchen, Germany. **Kumah, K. K.** Combining Microwave Links and Meteosat Second Generation Data for Rainfall Studies in Kenya. Short presentation.

KIT/HyMet CML Symposium 2019 – Garmisch-Partenkirchen, Germany. **Kumah, K. K.** Opportunistic Use of Commercial Microwave Links and Meteosat Second Generation Data for Rainfall Studies in Kenya. Poster presentation.

PROFESSIONAL SERVICE & EXPERT ROLES

Peer Reviewer – Scientific Journals: Journal of Hydrometeorology; Hydrology and Earth System Sciences; Earth System Science Data; Remote Sensing; Journal of Geophysical Research: Atmospheres.

Grant Proposal Reviewer: Swiss National Science Foundation (SNSF) – external expert reviewer for SmartSense proposal on high-resolution rainfall sensing in smart cities using commercial microwave links and acoustic sensors.

Professional Network / Working Group Membership: COST Action CA20136 SmartSense – approved Working Group 3 member in a European research network focused on opportunistic precipitation sensing and related environmental monitoring methods.

MEDIA COVERAGE & PUBLIC RECOGNITION

APS Physics Magazine – American Physical Society (2021). Featured in “Mobile Phone Signals Hold Real-Time Rainfall Information,” an independent science-news article on the use of mobile-phone network signals for rainfall monitoring.

myScience News / University of Twente ITC (2022). Featured in “ITC Algorithm Detects Rainfall in Rural Africa,” covering doctoral rainfall-estimation work, machine-learning methodology, and applications in data-sparse regions.

SKILLS

Programming & Scientific Computing: Python, NumPy, Pandas, Xarray, Dask, scikit-learn, Matplotlib, Cartopy, NetCDF/HDF5/GRIB/GeoTIFF, Git, Linux/HPC batch workflows, reproducible data pipelines.

Remote Sensing & GIS: AVHRR, MSG SEVIRI, MODIS, GPM IMERG, GRACE, Rasterio, Rioxarray, GeoPandas, GDAL, ArcGIS/QGIS, raster reprojection, large-scale resampling.

Machine Learning & Retrieval Methods: Random Forest, XGBoost, neural networks, feature engineering, cross-validation, uncertainty diagnostics, double-machine-learning retrieval strategies for satellite precipitation.

Data Products & Validation: Multi-sensor validation, bias correction, climate data records, PDF/CDF analysis, precipitation-product diagnostics, interannual variability, trend statistics, production-style processing workflows.